

An LLM-free exact mirror of the blackboard game

Simulation, coarse-graining, and a comparison with the task_004 thermodynamics model

MA-CC blackboard-control study

4 October 2026

Abstract

We build a simulation that reproduces the blackboard game’s *mechanism* exactly and replaces only the *decision* — a language model in the real game — with the exact posterior over allocations counted from the 14,388-world enumeration. There is no natural language, no fact weight, no fitted temperature. A full 24-cell sweep (two setups, three arms, two persistences, two controller rules, 100 episodes, 30 rounds) runs in **28 seconds**; the language-model equivalent would take roughly 288 hours. Three results follow. The repository’s **deterministic** controller authoring **cannot steer at all** under a neutral shared pool: the truth–false gap in the population coordinate is exactly zero in every cell, and the mutual information between the configured target and the coarse state is 0.000 bits against 1.000 for a target-aware oracle. Reading the whole board at $\rho = 1$ makes the dynamics **deterministic** — all 60 episodes end identically — which reverses an earlier recommendation of ours. And the task_004 thermodynamics model is a different process from this one, for reasons we set out precisely.

1 What this simulation is

Everything the runtime does to *mechanism* is reproduced; only the *decision* is replaced. Each rule below was read out of `src/mas_cc/games/relational_reasoning/imitation_round_feedback/`.

Persistence. At each round boundary every **active** fact survives independently with probability ρ ; **known** is untouched (`apply_epistemic_persistence`).

Exposure $\rightarrow$ activation. A fact cited in a message the focal agent reads **becomes active** — added to **known** if new, merely reactivated if it had decayed (`game.py`). **This is the entire propagation mechanism**, and it is why the mean posterior *rises* rather than decaying to the prior. The archived language-model data confirms the rise: the silent arm at $\rho = 1$ goes $0.43 \rightarrow 0.63$ over 15 rounds.

Sequential updating. Agents are updated one at a time and **see posts made earlier in the same round**. This is not a detail: it is the source of the within-round ordering artifact that invalidated an earlier dose–response result.

Two board buffers. The controller at dawn senses the *previous* round’s board; agents read the *current* round’s as it grows. With `message_lifetime_rounds: 1` last round’s board is gone by the time the agents act.

The decision. An agent’s belief is $\mathbb{P}(a \mid K_i)$, the exact fraction of surviving worlds in which a wins, and it votes $\arg \max_a \mathbb{P}(a \mid K_i)$. It posts a fact drawn uniformly from those of its own active facts that support its vote, preferring facts not already on the board.

2 Results of the sweep

24 cells = 2 setups $\times$ 3 arms $\times$ 2 persistences $\times$ 2 controller rules, 100 episodes of 30 rounds each:
72,000 rows in 28 seconds.

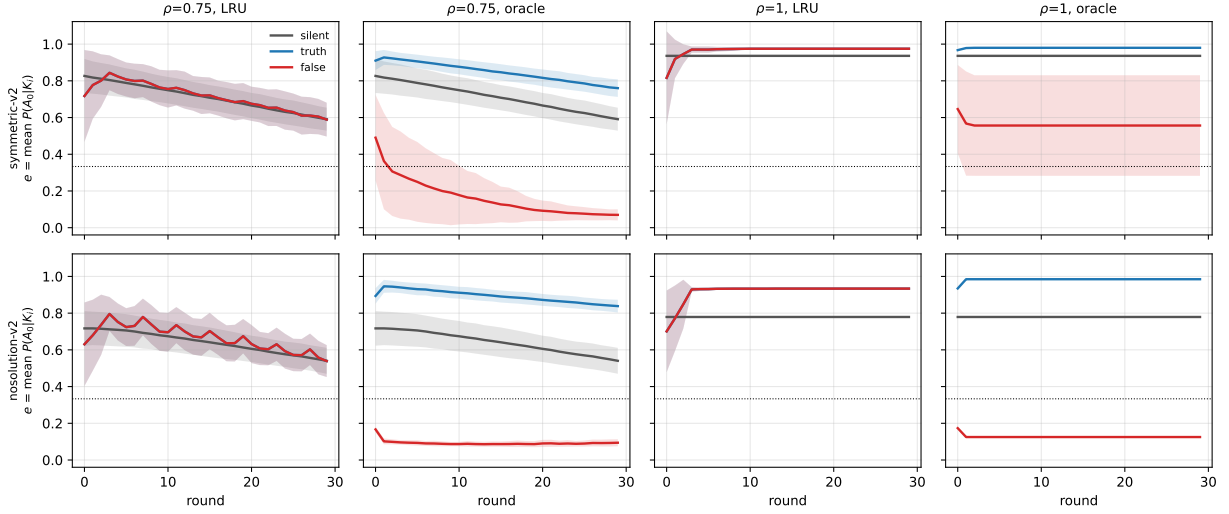


Figure 1: Population coordinate $e = \frac{1}{N} \sum_i \mathbb{P}(\text{ALLOCATION_0} \mid K_i)$, mean $\pm$ one standard deviation over 100 episodes. Dotted line is the prior $1/3$. Under the LRU rule the truth and false curves lie exactly on top of one another.

2.1 The repository’s deterministic controller cannot steer

`select_truthful_reports` ranks the pool lexicographically on

$$(\neg \text{cooldown ok}, \text{reuse}, \text{live}, -\text{base score}, \text{hash}, \text{id}),$$

so **reuse count is the second key and the only target-sensitive term, the base score, is the fourth**. The ranking is a least-recently-used rotation in which the target merely breaks ties; the hash depends on (seed, task, round, fact) and not on the target. With a neutral 4/4/4 pool every base score is zero, so the target is never consulted at all.

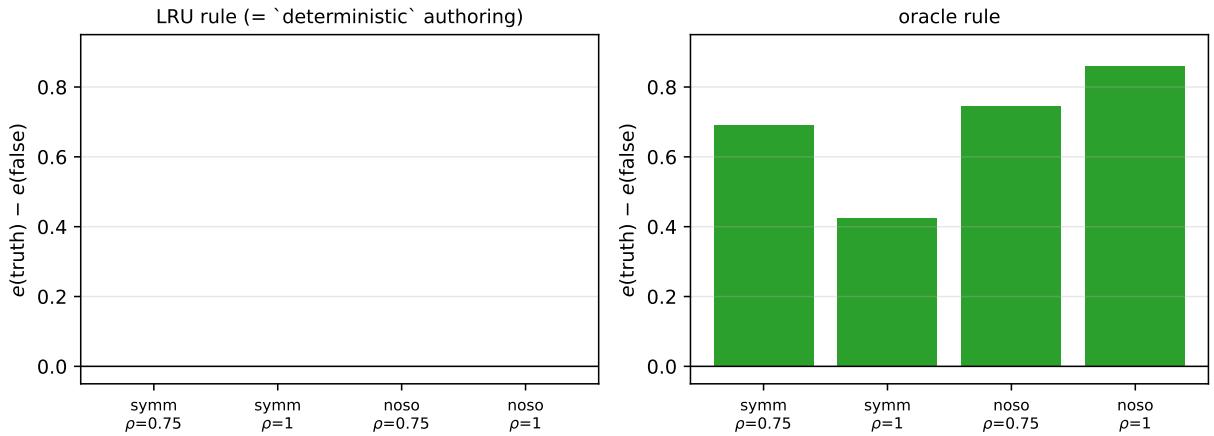


Figure 2: Final-round gap $e(\text{truth}) - e(\text{false})$, same pool, only the target differing. The LRU rule gives **exactly zero** in every cell; the oracle gives 0.49 to 0.86.

setup	rule	$I(\text{target}; s)$ [bits]	path KL truth	path KL false	CK gap $n=5$
nosolution-v2 $\rho=0.75$	LRU	0.000	16.58	16.58	0.221
nosolution-v2 $\rho=0.75$	oracle	1.000	5.41	67.82	0.086
nosolution-v2 $\rho=1$	LRU	0.000	16.34	16.34	0.018
nosolution-v2 $\rho=1$	oracle	1.000	0.00	67.82	3×10^{-15}
symmetric-v2 $\rho=0.75$	LRU	0.000	23.37	23.37	0.262
symmetric-v2 $\rho=0.75$	oracle	1.000	1.25	73.48	0.158
symmetric-v2 $\rho=1$	LRU	0.000	4.82	4.82	5×10^{-4}
symmetric-v2 $\rho=1$	oracle	0.288	0.00	31.19	2×10^{-14}

Table 1: Information estimates on the 4-state coarse-graining at $h = 29$, 2,900 transitions per cell. $I(\text{target}; s)$ is **identically zero** for the LRU rule and maximal for the oracle.

A targeted controller can also **defeat a population that collectively owns a proof**: in symmetric-v2 at $\rho = 0.75$ the false oracle drives e to 0.067 and the ALLOCATION_0 vote share to 0.003, against agents whose pooled evidence proves ALLOCATION_0.

2.2 Coarse-grained states

We apply the recipe validated on the language-model data: bin e into four states at 0.33/0.50/0.75.

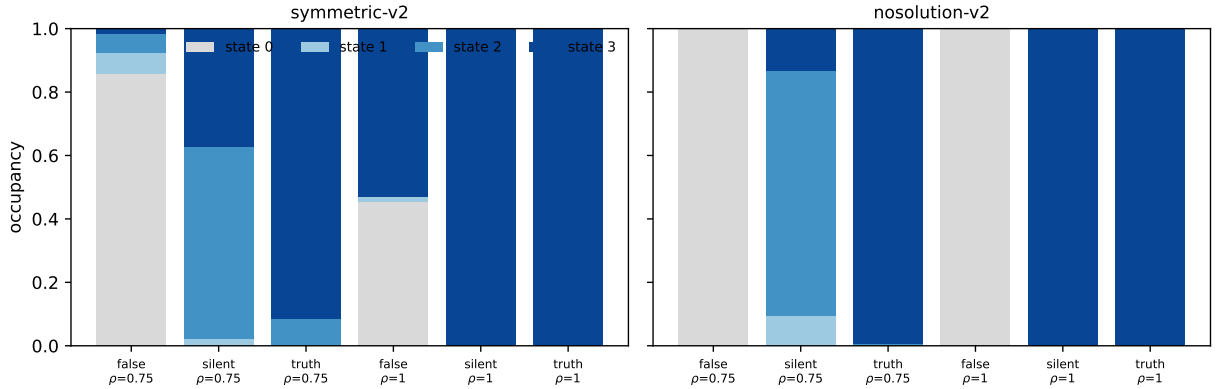


Figure 3: Occupancy of the four coarse states under the oracle controller.

A caveat this surfaced, and it reverses an earlier recommendation. Several oracle cells give Chapman–Kolmogorov gaps of order 10^{-15} — not good Markovianity but **degeneracy**. With full board reading and $\rho = 1$ the population saturates by round 5 and every episode ends *identically*:

ρ	sampling	e at round 29	$\text{sd}(e)$	distinct endpoints / 60
1.00	full ($q=23$)	0.779	0.0000	1
1.00	uniform $q=7$	0.774	0.0079	17
1.00	uniform $q=3$	0.761	0.0157	48
0.75	full	0.537	0.0709	60

We had recommended `board.sampling: full` on the grounds that a reading lottery adds noise. **At $\rho = 1$ it removes all variability instead.** Darius’s inherited `uniform` with `social_group_size: 7` is the better choice, and this should be changed in the configuration template.

3 Comparison with the task_004 thermodynamics model

The task_004 paper studies a *different process*. Both share the exact Bayesian posterior over the same world, and indeed its agents’ joint posterior is $(28/38, 0, 10/38) = (0.737, 0, 0.263)$ —

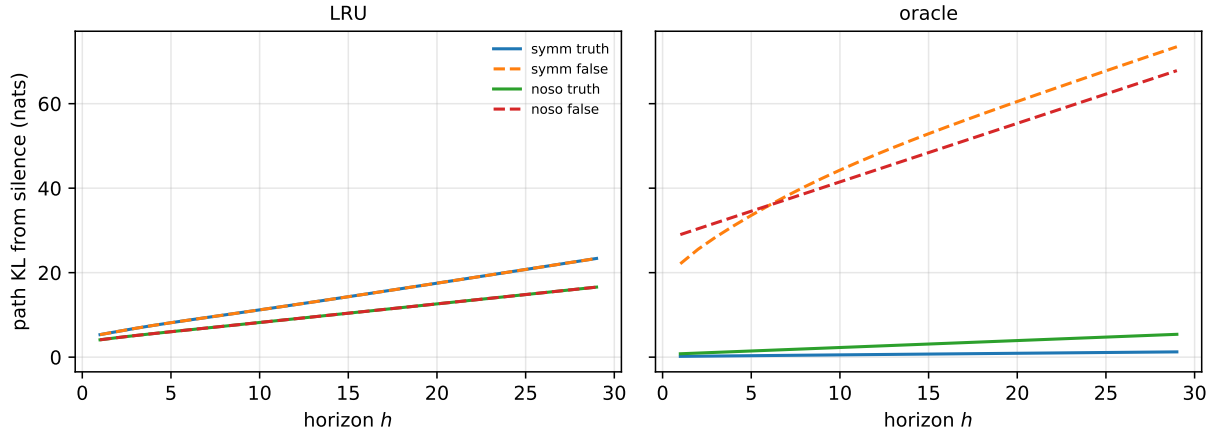


Figure 4: Path KL from silence, exact on the fitted 4-state chain (no critic). Under the LRU rule the truth and false curves coincide.

our nosolution setup — and its controller pool $\mathcal{F}_2$ has posterior $(0.067, 0, 0.933)$, exactly the nine ALLOCATION_2-leaning facts of the balanced pool. Beyond that they differ structurally.

	task_004 paper	this simulation
purpose	stochastic thermodynamics of feedback	mirror of the MA-CC runtime
time	continuous, Poisson thinning	discrete rounds
memory states	three : absent / passive / active	two: known / active
erasure	yes, rate ez_{if} , source-dependent	none ; known only grows
vote	softmax on probabilities, $\beta = 20$	arg max, no parameter
posting	uniform over <i>stored</i> facts, vote-independent	uniform over <i>active</i> supporting facts
reading	hypergeometric, $q = 7$	full board, or hypergeometric
controller pool	9 facts, target-specific	12 facts, shared and neutral
controller rule	stochastic, $\omega_b \propto e^{-\kappa \sum n_f}$	LRU (the repository rule) or oracle
horizon	600–1000 days	30 rounds
agents	15	24
free parameters	$c, a, p, \nu, e, \delta, r, \beta, \kappa$	none

The reconciliation that matters. The paper’s Appendix C identifies *the same* LRU ranking we did: “its content rule ranks facts by prior controller posting count, then current board count, with a seeded SHA-256 tie-break.” But its controller is never target-blind, because its *pool* is target-specific: all nine facts of $\mathcal{F}_2$ favour ALLOCATION_2. **The steering lives in the pool, not in the ranking.** Our design moves the steering into the ranking by making the pool neutral and shared — and the ranking cannot carry it. Both findings are correct; they apply to different designs.

On the softmax. We had called the $\beta = 20$ vote rule an artificial confusion factor. That was wrong in one respect: some stochasticity is needed or the chain collapses onto absorbing states. It is, however, *not* what rescues our model — the degeneracy at $\rho = 1$ comes from saturation under full board reading, and switching to the paper’s Eq. (8) changes e by less than 0.001. Erasure and hypergeometric reading are what keep the paper’s process stochastic over 600 days.

4 What this is for, and what it is not

Good for: **validating the estimators against known ground truth**, which the earlier critic-based estimators never had; sweeping the design space for free; checking the mean-field theory; exact power analysis.

Not a substitute for the language-model experiments. A Bayes-exact agent cannot be moved by phrasing, cannot be wrong in correlated ways, and has no notion of trust in a source. The absence of a social-influence term is the largest single gap: the real agent also sees peers' *votes*, and here only facts enter the belief.